

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

1. Solve the recurrence $T(n) = 5T(n/3) + n^2$.

Justify your answer showing how you derived your solution.

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2. Consider 4 employers $\{e_1, e_2, e_3, e_4\}$ and 4 applicants $\{a_1, a_2, a_3, a_4\}$.

preferences for employers					preferences for applicants						
e_1	:	2	3	4	1	a_1	:	3	2	1	4
e_2	:	3	2	1	4	a_2	:	3	2	1	4
e_3	:	3	2	1	4	a_3	:	1	3	4	2
e_4	:	3	2	4	1	a_4	:	3	4	2	1

Preferences are listed from high to low for employers and applicants.

- (a) Run the Gale-Shapley algorithm on the above input. Show all stages of the algorithm.
- (b) Describe what happens when the Gale-Shapley algorithm needs many rounds of offers to terminate. In particular, for which type of inputs does this happen?
3. In the graph $G = (V, E)$, with $V = \{0, 1, 2, 3, 4, 5\}$, and edges (i, j, w) , between i and j with distance w , in $E = \{(0, 3, 8), (0, 1, 5), (0, 2, 4), (1, 4, 1), (1, 2, 4), (2, 3, 2), (1, 3, 3), (3, 4, 7), (1, 5, 4), (2, 5, 2), (4, 5, 7)\}$, we want to compute all shortest paths from 0 to all other vertices.

- (a) Illustrate the execution of Dijkstra's algorithm on G .
Show all steps and the evolution of data structure to store the shortest paths.
- (b) Formulate the running time in function of the size of G . Justify your formula.
4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and the edge between i and j with weight w represented by the triple (i, j, w) in $E = \{(0, 5, 5), (1, 3, 8), (1, 2, 6), (2, 5, 5), (0, 2, 1), (3, 5, 4), (4, 5, 7), (3, 4, 5), (1, 4, 5)\}$.

- (a) Construct the minimum spanning tree with the reverse-delete algorithm. Show all steps.
- (b) For which type of graph would you not use the reverse-delete algorithm and use Prim or Kruskal? Justify your answer by referring to the asymptotic cost of the algorithms.
5. Let $P = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (2, 3), (3, 4)\}$.

- (a) Illustrate the application of a divide-and-conquer algorithm to compute the closest pair of points in P .
- (b) Use the example to explain why the asymptotic cost is $O(n \log(n))$ for n points.
6. Let $C = A \times B$, with $A = [1, 0, 1, 0]$ and $B = [0, 0, 0, 1]$ given as lists of bits, in increasing order of significance, that is: lowest bit first, highest bit last.

- (a) Apply Karatsuba's algorithm to compute C . Show all steps.
- (b) Give the recurrence relation which justifies the running time.
Explain the recurrence by referring to the example.

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